

TEST PLAN DOCUMENT

Project: Notes Hybrid Automation Framework

Application Under Test: Notes App – ExpandTesting

Version: 1.0

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Role: Automation Test Engineer

1. Introduction

This Test Plan document defines the testing strategy, scope, objectives, environment, execution flow, and deliverables for validating the Notes Application using a Hybrid Automation Framework. The framework includes UI automation, API automation, Hybrid testing, MCP-inspired execution, CI/CD integration, self-healing concepts, and Allure reporting.

2. Objectives

- Validate login functionality.
- Validate note creation and deletion functionality.
- Validate backend APIs using Rest Assured.
- Verify UI and API synchronization.
- Validate MCP command-driven execution.
- Validate negative scenarios.
- Verify application performance and execution stability.
- Generate professional reports using Allure.

3. Scope

- UI Testing
- API Testing
- Hybrid End-to-End Testing
- Negative Testing
- Performance Validation
- MCP-inspired Dynamic Execution
- CI/CD Validation using Jenkins

4. Functional Requirements Covered

- FR-01 User Login Validation
- FR-02 Create Notes Functionality
- FR-03 Delete Notes Functionality
- FR-04 Notes API Validation
- FR-05 UI and API Synchronization
- FR-06 MCP Dynamic Command Execution
- FR-07 Negative Scenario Validation
- FR-08 Performance Validation

5. Test Strategy

- UI testing performed using Selenium WebDriver.
- API testing performed using Rest Assured.
- Hybrid testing validates UI and API synchronization.
- MCP layer validates dynamic command-driven execution.
- Self-healing mechanism improves locator stability.
- TestNG manages execution and parallel execution.
- Jenkins executes CI/CD pipeline automatically.

6. Test Environment

- Operating System: Windows 10/11
- Programming Language: Java
- UI Automation Tool: Selenium WebDriver
- API Automation Tool: Rest Assured
- Testing Framework: TestNG
- Build Tool: Maven
- CI/CD Tool: Jenkins
- Reporting Tool: Allure Reports
- IDE: IntelliJ IDEA

7. Test Data

- Valid Email
- Invalid Email
- Valid Password
- Invalid Password
- Note Title
- Note Description

8. Entry Criteria

- Application should be accessible.
- APIs should be available.
- Test environment should be configured.
- Test data should be prepared.

9. Exit Criteria

- All test cases executed.
- Critical defects resolved.
- Reports generated successfully.
- RTM completed.

10. Deliverables

- Test Plan
- Test Scenarios
- Test Cases
- RTM
- Execution Report
- Allure Reports
- Jenkins Build Reports

11. Roles and Responsibilities

- QA Engineer – Execute and validate test cases
- Automation Engineer – Develop framework

- Developer – Fix defects
- Reviewer – Review testing artifacts

12. Risks and Mitigation

- Application downtime may delay testing.
- Environment instability may block execution.
- Locator changes may affect automation.
- Self-healing mechanism reduces locator failures.

13. Defect Management Process

- Defect ID
- Summary
- Steps to Reproduce
- Expected Result
- Actual Result
- Severity
- Priority
- Screenshots

14. Test Execution Summary

- Total Test Cases: 21
- Passed: 21
- Failed: 0
- Execution Status: SUCCESS

15. Approval

- QA Reviewer – Pending
- Project Reviewer – Pending